Cape Peninsula University of Technology

APPLICATIONS DEVELOPMENT THEORY 4

SUBJECT GUIDE FOR:	2025
COURSE CODE:	APT470S

Class Exercise: Group Work

Instructions:

i. *Total Marks:* ***50***

ii. *Attempt all questions.*

iii. *This is a* ***group assignment****, and the solution must be uploaded by* ***only*** *the group leader.*

iv. *Solutions must be uploaded to Blackboard by midnight today* ***April 08, 2026****.*

1. **Scenario:** A **smart washing machine** operates through several states depending on user input and internal conditions. The machine starts in an *Idle* state, waiting for user action. When the user closes the door and selects a wash program, the machine transitions to *Ready*. Once the user presses *Start*, the machine moves to *the Washing mode*. During washing, the user can press *Pause* to temporarily *halt* the *operation* or *Cancel* to stop completely. After the wash finishes, the machine transitions to *Rinsing*, then *Spinning*, and finally *Complete*. If the door is opened at any time before completion, the machine returns to *Idle*.

Question

Draw a ***UML State Machine Diagram*** *for the smart washing machine described above. Show all states, transitions, and actions. Include entry and exit actions where appropriate. [20 Marks]*

2. **Scenario:** A university's student management system needs to transfer student grade data from the **Faculty Database** to the **Central Academic Records System (CARS)**. The process involves the following actors:

- **Faculty Admin** (initiates the transfer)
- **StudentInfo Module** (handles student data locally)
- **Faculty-DB** (faculty-level database)
- **Authorization Service** (validates transfer rights)
- **CARS** (central repository for academic records)

Process:

i. The **Faculty Admin** logs into the system.

ii. Under an **alt fragment**, two possible actions occur:
   - **[sendGrades]**: Admin requests to transfer updated grade information.
   - **[sendTranscript]**: Admin requests to transfer a summarized transcript.

iii. In the **sendGrades** path:
   - Admin triggers updateGrades() in **StudentInfo Module**.
   - This calls pushGrades(StudentID) to **Faculty-DB**.
   - **Faculty-DB** requests authorization via authorize(UserRole, StudentID) from **Authorization Service**.
   - Once authorized, **Faculty-DB** sends updateRecord(StudentID) to **CARS**.
   - **CARS** confirms with update OK.
   - A success message is returned to the **Faculty Admin**.

iv. In the **sendTranscript** path:
   - Admin triggers generateTranscript() in **StudentInfo Module**.
   - These calls summarize (StudentID) to **Faculty-DB**.
   - Authorization is checked as above.
   - **Faculty-DB** sends updateTranscript(StudentID) to **CARS**.
   - **CARS** confirms with update OK.
   - A success message is returned to the **Faculty Admin**.

v. Finally, the **Faculty Admin** logs out.

Question:

Using the scenario above, draw a UML sequence diagram that illustrates the interactions between the Faculty Admin, StudentInfo Module, Faculty-DB, Authorization Service, and CARS for both the [sendGrades] and [sendTranscript] paths. Clearly show the alt fragment and the message flow. [30 Marks]